

	mov a,#0c0h acall command ;Force cursor at beginning of the second line mov dptr,#msg4 acall disp_msg d3_loop: mov a,#0c9h acall command ;Force cursor at beginning of the second line acall disp_ss acall 1_delay DJNZ b,d3_loop setb BUZZER acall 1_delay clr BUZZER ajmp control_loop disp_ss: push b MOV 30h,b acall hexbcd mov a,34h subb a,#06h acall hexasci acall disp_data mov a,33h acall hexasci acall disp_data pop b ret hexasci: add A,#36h jnb PSW.6,skip add A,#06h skip: subb a,#06h ret hexbcd: MOV 35h,#0 MOV 36h,#0 MOV 37h,#0 MOV A,30h MOV B,#10 DIV AB MOV 33h,B MOV 34h,A MOV A,31h JZ go_next MOV B,#10 DIV AB MOV 35h,B MOV 36h,A ; Value in 35h is 256 times higher than value standing in 33h, 34h. Mply 35h with 6, ; add to 33h, mply 35h with 5, add to 34h, mply 35h with 2 and replace itself in 35h MOV R0,#35h ; Point to 35h ACALL TRANSF ; Transform 35h to 3 lower position INC R0 ; Repeat from 36h, now all registers are 1 higher ACALL TRANSF ; Transform 36h to 3 higher positions go_next: MOV R0,#33h ; Point to lowest digit i.e. first digit ACALL DECAJ RET TRANSF: MOV A,R0 MOV B,#6 MUL AB DEC R0 DEC R0 ADD A,R0 MOV R0,A INC R0 INC R0 MOV A,R0 MOV B,#5 MUL AB DEC R0 ADD A,R0 DECLOP: MOV B,#10 DIV AB MOV R0,B INC R0 ADD A,R0 MOV R0,A CJNE R0,#37h,DECLOP RET command: push dph push dpl mov _data,a acall delay ;Write when when display not busy clr RS ;Select command regis- ter setb EN ;Set Enable terminal of LCD nop nop clr EN pop dpl pop dph ret disp_data: push dph push dpl acall delay mov _data,a ;Get data for dis- play setb RS ;Select data register setb EN ;Enable LCD (Strobe) LCD) nop nop clr EN ;Clear Enable and latch data pop dpl pop dph ret disp_msg:acall delay clr a movc a,#a+dptr jz exit acall disp_data inc dptr sjmp disp_msg exit: ret set_time: mov a,#01h acall command ;Clear LCD acall 1_delay jnb SW1,\$ acall 1_delay acall s_puk jnb SW2,\$ acall 1_delay acall s_kum jnb SW3,\$ acall s_rech acall 1_delay call control_loop ret s_puk: acall s_delay mov a,#80h acall command ;Force cursor at beginning of the second line mov dptr,#msg5 acall disp_msg acall 1_delay inc1: mov a, 40h add a, #1 mov 40h,a cjne a,#00AH,max1 mov 40h,#1 mov b, 40h max1: mov a,#0c9h acall command acall disp_ss acall 1_delay jnb SW1, inc1 ret s_kum: acall s_delay mov a,#80h acall command ;Force cursor at beginning of the second line mov dptr,#msg6 acall disp_msg acall 1_delay inc2: mov a, 41h add a, #1 mov 41h,a cjne a,#03CH,max2 mov 41h,#1 max2: mov b, 41h mov a,#0c9h acall command acall disp_ss acall 1_delay jnb SW2, inc2 ret s_rech: acall s_delay mov a,#80h acall command ;Force cursor at beginning of the second line mov dptr,#msg7 acall disp_msg acall 1_delay inc3: mov a, 42h add a, #1 mov 42h,a cjne a,#14H,max3 mov 42h,#1 max3: mov b, 42h mov a,#0c9h acall command acall disp_ss acall 1_delay jnb SW3, inc3 ret l_delay: mov R4,#18h loop3: acall delay djnz R4,loop3 ret delay: mov r7,#50h loop2: acall s_delay djnz r7,loop2 ret s_delay: mov R6,#0ffh loop1: djnz R6,loop1 ret msg1: db 'FRANAYAM TIMER',0 msg2: db 'PURAK: ',0 msg3: db 'KUMBHAK: ',0 msg4: db 'RECHAK: ',0 msg5: db 'SET PURAK TIME',0 msg6: db 'SET KUMBHAK TIME',0 msg7: db 'SET RECHAK TIME',0 END
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